

Professional Ballistics Manual

1 Introduction

This manual presents a complete modern framework for exterior ballistics, including:

- Point-mass and 6-DOF models
- Doppler-based drag modeling
- Numerical integration methods
- Practical shooting outputs (DOPE)

2 Equations of Motion

$$\begin{aligned}\dot{\mathbf{r}} &= \mathbf{v} \\ \dot{\mathbf{v}} &= \frac{1}{m}(F_{drag} + F_{gravity} + F_{coriolis})\end{aligned}$$

3 Drag Model

$$F_D = -\frac{1}{2}\rho v^2 C_D(M)\hat{\mathbf{v}}$$

4 6-DOF Model

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}(\mathbf{M} - \boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega})$$

5 Doppler Data

Drag coefficients are obtained from radar data and interpolated via splines.

6 DOPE

$$MIL = \frac{\Delta y}{x} \times 1000$$

7 Conclusion

This document provides a full pipeline from physics to practical shooting solutions.